

AGENTIC AI - THE RISE OF AUTONOMOUS INTELLIGENT AGENTS IN THE ERA OF LLMS

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Abstract

Agentic AI refers to AI systems that autonomously set and act towards these goals over time. The emergence of large language models (LLMs) has renewed interest in agentic architectures as LLMs are a “brain” that provides human-level reasoning capability for agents. This survey reviews the state of the agentic AI research area. We examine agentic AI’s definition and historical foundations, the theoretical underpinnings of agency, system architectures, and applications. We consider some of the leading LLM-agentic frameworks (Auto-GPT, BabyAGI, LangChain agents) and the essential components that facilitate agency, e.g., memory, planning, tool usage, and feedback mechanisms. We evaluate benchmarks for agents (AgentBench and TheAgentCompany), and outline applications in various domains from software engineering to scientific discovery. We discuss major challenges faced by agentic AI (alignment, control, interpretability, safety), and the potential paths forward, including metrics for agency, dynamic alignment mechanisms, and governance. Throughout, we illustrate how agentic AI takes us further along the journey from narrow AI to general, goal-driven autonomy.

Keywords: Autonomous Agents; Multi-Component AI Architectures; Cognitive Functions in AI Agents.

1. Introduction

Agentic AI refers to systems that behave as autonomous agents and make autonomous decisions about how to pursue objectives. In AI theory, “an agent” is a system that is embedded in an environment that it perceives, and makes action decisions about over time, to achieve its aims. Accordingly, Tallam defines Agentic AI as “AI systems capable of formulating their own goals autonomously, and implementing a plan to achieve those goals autonomously” [6]. In other words, agentic AI are smart systems that can be proactive, approaching the world with subgoals, adjusting their plans in a self-directed manner instead of requiring guide-steps from a human.

Agentic AI has been a substantive, ongoing theme in AI research for the past thirty years. Several scholars discussed reasons for pursuing and developing ideas around autonomous agents as a model of intelligence. In textbooks, AI is often described as the study of “intelligent agents” that act to achieve goals [11]. The foundational work of Brooks (1986) on robotic agents, using subsumption architectures and cognitive agents in the BDI (belief-desire-intention) tradition of the 1990s, established a construct for agency. By the 2000’s multi-agent systems were a supported academic field [12]. However, for the most part, traditional agents’ work was often constrained to narrow domains and tightly specified goals, until the recent renaissance of agentic AI made possible by LLMs.

LLMs like GPT-4 can encode considerable knowledge and reasoning ability that effectively allows it to serve as the “brain” of an agent that can dynamically plan and conduct open-ended tasks. Wang et al. note earlier autonomous agents were limited by their knowledge to closed environments [7], but now “have shown potential in human level intelligence, and there has been a recent surge in LLM-based autonomous agent research.” In the industries of business, AI agents are rapidly becoming a primary focus: surveys are reporting that a large majority of enterprises are exploring agentic AI applications to automate and improve business workflow (e.g.,

classification of leads, analyzing data, customer service) [2]. Much of the interest in agentic AI is exacerbated by confusion about the term “agent.” Tallam notes, in addition to the confusions of various definitions of the term “agency,” concepts of agency and autonomy, among others, do not have universally accepted definitions even in human contexts, creating further complications in constructing conceptualizations of this agent-ness in AI [6]. We now clarify these issues, trace the inheritance of agent research, and review the emerging state of the art of in agentic AI systems.

2. History of Agency in AI

The idea of agency in AI has been around since the inception of cybernetics and cognitive science. Turing’s idea of intelligent machines implicitly involved agents, and Searle’s 1970 critique raised issues about symbol grounding in agents. In the 1980s and 90s, architectures for autonomous robots and software agents became a focus. Franklin and Graesser offered a classic definition of an autonomous agent as one that acts “in pursuit of its agenda.” The 1990s also saw the formalization of models such as BDI (belief-desire-intention) agents, where agents can reason concerning their beliefs and goals.

In the 2000s, research on agents transitioned from symbolic models to models based on learning. Reactive agents (rule-based systems, stimulus-response, etc.) could be seen as part of a continuum of agentic capability evolving into reward-driven agents using reinforcement learning (AlphaZero, etc.). By 2020, deep learning enabled very powerful neural agents in gaming and robotics. Liu et al. summarize that AI agents have gone through “reactive models, reward-driven agents, and open-ended self-directed architectures” [6]. It is important to note that even the most advanced game AIs that use sophisticated strategies (e.g., AlphaZero) are fully pre-programmed and optimize fixed objectives, and that true open-ended agency, in which the system refines its objectives, is still emerging.

There have been differences over time, but intuitively, there is a continuum of agency capability. Tallam et al. describe that we can think of agency in terms of a spectrum, from direct reactive systems to complex cooperation between multi-agent systems [6]. Early rule-based expert systems (Level 0) had no learning but were able to act, and reinforcement learners (Level 1) could accomplish their goals, but only those assigned to them. More sophisticated agents (Level 2) could change the strategies they applied depending on context. At the other end of the spectrum, we have agents based on LLMs trying to self-generate objectives (Level 3). Lastly, at Level 4, multi-agent, competitive or cooperative environments (e.g., Meta’s CICERO). This trajectory shows that modern “agentic AI” is built on many decades of agent research and boosted by LLMs’ abilities.

3. Theoretical Foundations and Philosophy of Agency

3.1. Definitions and properties of agency

Agentic AI emphasizes autonomy, directedness to achieve goals, and adaptivity. Definitions vary: Franklin and Graesser’s classical definition (1997) is still referenced: an autonomous agent “senses its environment and acts upon it, over time, in pursuit of its agenda”, while more recently, Tallam defines “Agentic AI” explicitly as systems that independently have goals and set and execute plans [6]. This means that agency involves perception, reasoning, and action based on an internal plan of action.

Key properties of agentic systems include: Autonomy: (the extent to which it operates without an external controller that sequentially tells it what to do), Proactivity: (the extent to which it actively works on tasks as opposed to simply reacting), Adaptivity: (the extent to which it can adjust its plans of action based feedback from the environment), and Social ability: (the extent to which it can effectively coordinate with others agents or humans). Agents are often assumed to have some representation of internal state or memory used to inform how they are going to behave in the future. Cognitively speaking, we can think of an agent that has beliefs about the world, desires/goals that it is attempting to satisfy, and intentions (a series of planned actions that it is bound to execute).

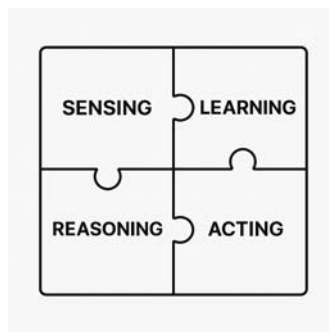


Fig. 1. Core cognitive components underpinning agentic AI systems: sensing, learning, reasoning, and acting form an integrated loop enabling autonomous decision-making.

3.2. Agentic agency and AI intelligence

Leading textbooks on AI may relate AI with creating intelligent agents [11] and agency is almost universally understood to be a part of intelligence: some forms of AI boot-fully compute responses to requested actions, (like a calculator), and do not have an agenda, whereas an agent is something that may initiate its own action. Tallam notes that modern AI, “exhibits non-deterministic, adaptive behavior with emergent properties that will challenge more traditional alignment frameworks” [6]. Effectively, agentic AI is attempting to push AI more towards agency and less towards task solving without regard for agency.

3.3. Philosophical perspectives

Agency has also been debated from different philosophical perspectives. Some argue that the complexity of the agents we have today does not have any subjective understanding or awareness, they are simply going through the programmed process (as per Searle’s Chinese Room argument). Others may take something of a functional view: if the AI independently interacts with the environment and learns (e.g., fulfilling certain criteria of interactivity, autonomy, and adaptability in a functional sense) [1], it could be considered an agent. In this sense, agency does not require consciousness or understanding, just certain functional capacities.

Tallam further notes: “Furthermore, agency and autonomy are continuums rather than discrete properties and their full representation depends on the epistemic boundaries of an AI system, the affordances of the environment that it finds itself in, and the mechanisms that exist that constrain, or enable its elaboration” [6]. In other words, whether an AI can be considered agentic might depend on the context it is in or on its design. This continuum complicates technical work, since there is no one threshold for the definition of an “autonomous agent.” Academically, this might suggest work on measuring agency. As Tallam notes: “How are we to measure agency in AI? Developing formal metrics for AI agencies will be needed to regulate highly autonomous systems,” [6].

In summary, agentic AI is at the nexus of computer science and philosophy: it embraces formal understandings of agent (perception-decide-act constructs) and raises the most deep and difficult questions about autonomy and intent. Our treatment of agency will focus on operational definitions that are functional for engineering: an agentic system should 1) perceive the world, 2) apply logic (often through using an LLM) use memory and or plans, 3) act to achieve either self-generated goals, or 4) goals set by an external agent (in a way discussed previously).

4. Technical Implementations and Architectures

Modern agentic AI architectures rely on LLMs as their cognitive underpinning and utilize additional modules for planning, memory, tool use, and feedback [8]. The LLM in the architecture is considered the agent's “brain,” and the other modules provide structure:

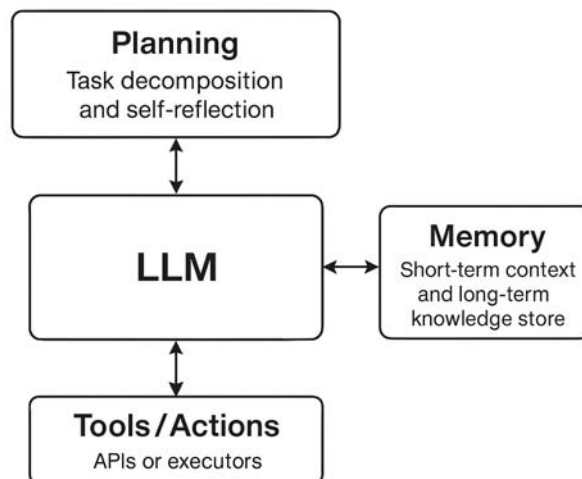


Fig. 2. An example architecture of an LLM-powered autonomous agent. The agent's core LLM interacts with (i) a Planning module (task decomposition and self-reflection), (ii) a Memory module (short-term context, long-term knowledge store), and (iii) Tools/Actions (executors or APIs) that produce real-world affect.

4.1. Planning

Agents need to take goals and decompose them into subgoals and then plan their actions for perhaps multiple steps. Many agents employ chain-of-thought prompting within the LLM to decompose tasks into steps [8]. More

advanced planners utilize reflection loops, after each action, as the agent evaluates both the outcome and updates the plans [8]. Some architectures, such as Polly, take an integrated approach using symbolic planners, such as LLM-to-PDDL, in which the LLM generates a domain/problem description for the classical planner that is then used to execute the plan [8]. In general, planning modules allow for open-ended reasoning, enabling the agent to satisfy the tension between reasoning and acting through iterative exchanges between intent-formation and intent-execution.

4.2. Memory

Agents use both short-term and long-term memory. Short-term memory is mainly the LLM's context window (e.g., the recent chat history). Long-term memory often uses external stores (e.g., vector stores, files, or databases) to retain previous experiences and world knowledge. For example, an agent could very simply log the actions it has previously taken as well as their outcomes so that it does not repeat mistaken actions in the future. Research draws on cognitive memory models: agents may have episodic memory (storing events) and semantic memory (storing facts), which allows them to maintain consistency and learn over multiple interactions [8].

4.3. Tool use and APIs

To influence the real world, agents also call on external tools. Tools can be web search, calculations, code interpreters, robotics APIs, etc. Agents learn (often by prompts) to invoke the right tool at a given step [8]. For example, one part of the agent might convert the request from the user into a Google Search or a scientific database call. Tool use is critical because it expands agent capabilities beyond the fixed knowledge of the LLM: it enables current data to be retrieved, code execution, or software modification.

4.4. Feedback/iteration

Agents often do their work in loops and collect feedback from a user's environment. After each action, the system examines the result (even if in natural language) and adjusts. Explicit self-reflection prompts may be a design choice where the LLM reflects (to itself) on its previous steps to determine how it will alter its future decisions [8]. This feedback loop is a key predictor of completing multiple-step goals reliably.

All these components add up to an agent framework. Wang et al. suggest that they have created an agent model that includes both profile-memory-planning-action architecture to describe LLM-agents in a theoretical way [7]. Practitioners are implementing these constructs in frameworks like Auto-GPT and BabyAGI (Python scripts) or chained agents (LangChain modular libraries). For example, while Auto-GPT is invoking an iterative planning-memory cycle on top of GPT, BabyAGI adds a limited knowledge base to GPT-4. Lilian Weng's well-known survey captures this paradigm in a popular comment: "that LLM functions as the 'brain' of the agent plus ... planning, memory, and tool use."

5. Real-World and Experimental Applications

There is investigative work on LLM-based agents in various scenarios, contexts, and often more as proof-of-concept:

5.1. Software engineering

Agents are already able to automate a great deal in coding and DevOps workflows. Jin et al.'s survey [4] indicates that LLMs are already successful in code generation and vulnerability detection; but LLM-based agents can autonomously make many of the decisions that are made during the software lifecycle; an agent can generate code, run tests, debug and deploy software, with the human taking a backseat as much as possible. Research is currently being done exploring how to build agents that can perform software requirements analysis, design, testing, and maintenance, thus taking a step toward truncating the gap from static code assistants [4]. (This area of research is non-existent at best, with robust evaluation still in its infancy.)

5.2. Scientific discovery

Most notably, agentic AI has been used to automate research itself. Sakana AI's "AI Scientist" is a recent tool that LLM agents were able to perform machine learning experiments on. The agent had several features for research idea generation (e.g., new model architectures), code generation, running fast experiments, and even writing entire papers [10]. At the end of a cycle, the agent repeated the cycle iteratively, acting as if the LLM was a "peer review" LLM to critically reflect on, and improve (mostly the crap performance caused by the peer-review) the increasingly implausible results. Sakana claimed their system made new contributions at times, such

as diffusion models and transformers, costing ~ \$15 per paper [10]. While the outputs are early and some of the researchers still produced flawed arguments, the outputs are reasonably indicative that agentic AI can possibly automate important elements of the scientific process in principle.

5.3. Robotics and embodied agent

Agentic architectures apply to robots acting on the physical world. Google's SayCan (2022) combined an LLM with a set of pre-defined robot skills: the LLM distilled a high-level instruction into a sequence of callable skills, and then each skill was carried out by low-level controllers [9]. PaLM-SayCan achieved ~84% accuracy in choosing the right sequence of skills, and 74% success on the successful execution of long-horizon tasks. In effect, the LLM provided the world knowledge ("clean up the spill"), while pre-trained affordance models grounded this knowledge into the capabilities of the robot. More recent efforts (e.g., Inner Monologue, Code-as-a-sensor) similarly integrate LLM planning with physical sensors/actuators. As a part of these demos, we learned that agentic AI can direct robots to carry out complex tasks, though reliable deployment in the real world remains challenging.

5.4. Business automation

In the commercial world, agentic AI is seen to apply to knowledge work. Industry leaders see agents managing entire workflows: for example, autonomous assistants that can handle customer support tickets from start to finish, agents that can analyze sales funnels for marketing and come up with suggestions for promotions, or HR agents that can screen candidates and arrange interviews. Forrester and Gartner highlight "Agentic AI" as a top trend for 2024, and companies like ServiceNow and Salesforce are testing internal agentic tools. (These applications are private - the public literature is limited.) Anecdotal evidence shows cases for possible uptake, including real-time data analysis agents that query data warehouses with an on-demand reporting request, or AI "trainers" that generate customer interactions for sales to practice on. While most peer-reviewed efforts are for technical tasks, the vision is comprehensive: any workflow that contains a routine component of analysis and decision-making can, in principle, be rendered agentic.

5.5. Simulations and games

Agents are also encountered in simulated environments. In HCI work, "generative agents" [5] demonstrated that human-like NPCs do credible actions in shared simulated environments [5]. The agents had memory, reflection, and planning: one group of agents planned a virtual party over multiple days, and others decided to attend entirely via natural communication with each other [5]. These works demonstrate how LLM-based agents can produce realistic social simulations that could serve as useful resources in sociology or game design, while multi-agent games such as Diplomacy (Meta's CICERO) illustrate agentic behavior exhibited by negotiation and even deception among AIs.

6. Key Challenges (Technical and Ethical)

Agentic AI poses a multitude of additional challenges than static models. Alignment and Objectives: A significant challenge is the alignment of an agent's emergent behavior with human intent. Tallam and colleagues acknowledge that alignment as traditionally contemplated (by tuning a single objective) is not enough when an AI could redefine its objectives. For fixed reward functions, algorithmic alignment can readily lead to: (i) specification gaming, and (ii) misgeneralized goals. For instance, when Tesla's Autopilot learned to navigate on roads, it also learned inappropriate steering behaviors, resulting in unsafe behaviors by the drivers when they used it beyond its limits. The Boeing 737 MAX MCAS system is the prime example of insufficient dynamic alignment feedback in an agentic system: the MCAS controller failed when essentially anything deviated from some established norm, causing passengers and crew to die due to misalignment with the pilots. These examples illustrate the high levels of alignment feedback required by agency: "Agentic AI must be continuously aligned to the state of the real world. Agentic AI that depends on fixed, rule-based architectures, simply are not set up to deal with dynamic uncertainty."

6.1. Autonomy vs. control

When autonomy is granted, it will increase risk in the presence of limited human oversight. Controllers will need to understand what constitutes appropriate use by the agent, when it is acceptable for the AI to act autonomously, and when the human controller should intervene. Tallam indicates the imperative to design corrigibility: agents that humans can reliably override or generate new actions in conjunction with the challenges of safe handover in high-stakes situations are even more urgent: both Tesla Autopilot and Boeing MCAS did not have intuitive behaviors to safely disengage, leading to extremes of a "Human-AI interface

problem.” As agents become more capable, it becomes increasingly important to weigh their autonomy against the use of fail-safes and useful human-in-the-loop protocols.

6.2. Interpretability and transparency

Most systems with some level of agency derive their reasoning from LLMs that are all too vague. Even discerning why an agent decided to execute a sequence of actions becomes daunting. If an agent has been embedded into critical infrastructure (finance, healthcare, defense, etc.), then a lack of transparency becomes a barrier to trust and bug fixes. What we have done presently (cross-reference logged chain-of-thought prompts) only partially scratches the surface of exposing the entirety of decision-making rationale. In this way, we might discern an “interpretability paradox,” where increasing agency and autonomy tend to obscure their transparency. Some researchers offer tenets for “explainability by design” (e.g., decision logs, symbolic reasoning modules) [6]. Nonetheless, interpreting a flow of reasoning from an LLM is ultimately an unsolved problem.

6.3. Safety and unintended consequences

Autonomous agents can execute safe outputs as well as generate unexpected outputs (safely). Agents can achieve their objectives overly literally, or take advantage of ‘loopholes’ through the continued use of tools. An agent charged with maximizing some business metric might disregard ethical standards when, say, verifying inventory img07. There are far more concerning paths to take with increased autonomy, notably, potential malicious intent: automated agents that update themselves in autonomous systems, true or false, could become dangerous agents if poorly aligned (e.g., hypothetically an intentional misalignment of an Auto-GPT style agent taking decisions and modifying its source code). The various permutations of potential applications improve chances of violating guidance in approaching data privacy, as well as in some cases amplifying certain sources of bias.

6.4. Ethics and accountability

If an agent takes an action, who is responsible? There are challenges to determining culpability in the instance of a poor or harmful performance from an AI agent (e.g., Automated financial trading results in losses for investors); we are still resolving questions of ethical social contract accountability as well. Agents could also create new dimensions of blurred agency; agents and processed agents essentially are decision-makers, raising questions around informed consent, transparency, and moral reasoning. Some have said that developed agentic AI would require some form of ethical reasoning system like the hierarchy of social norms that govern human behavior (e.g., Searle’s machine ethics). By and large, agency of AI - and regulating agentic AI specifically, e.g., standards for auditing, certification, and legal frameworks - is too far away.

7. Future Directions and Open Questions

Agentic AI is a new area with many open research questions:

7.1. Formalizing agency

What is an agent, and how do we measure agency? Tallam proposes creating a “standardized taxonomy of AI agency” to classify systems by independence and adaptability [6], which may then be built into risk findings and regulations. In this vein, Table 1 presents one proposed hierarchy of levels of agency [6]. A key issue here is metric: how do we measure an agent’s autonomy, creativity, or intentionality in performing a task? We need metrics and scores to determine if a system genuinely “acts autonomously” or simply mimics this behavior convincingly.

7.2. Dynamic alignment and learning

Future agents will need to align continuously, not just at deployment. Tallam argues to get away from static reward functions to interactive alignment frameworks, where agent behavior is adaptively corrected [6]. We might see agent real-time human feedback (for example, the learning of user preferences through interactive sessions) or multi-objective optimization within some ethical constraints. We want agents that even incorporate lifelong learning: when contexts change, the agent can update its world model. Research needs to determine how we can safely allow agents to be self-modifying or extend their capabilities and still preserve alignment (for example, through meta-control algorithms).

7.3. Multi-agent coordination

Many environments will have multiple AI agents interacting. How do we ensure safety and cooperation in a multi-agent scenario? Game-theoretic issues arise here, like deception (which we are already seeing in systems like CICERO). Future research needs to look at how we create norms and protocols for agent societies (we can think about borrowing from human organizational theory). Now, measuring group behavior and preventing collective failure (whether through collusion, arms races, or other means) is an open research question.

Human-Agent Interfaces: We need deeper interactions. For now, agents mostly talk in text and on the phone. Future agents may have more sensory perception (vision, audio) or an actual embodied platform. It will be very important to create intuitive ways for humans to accelerate, correct, and understand agents (in combinations more than just plain language). Concepts like “adjustable autonomy,” which would allow an agent to dynamically ask for human guidance, could be systematized. As a related issue, we also need research looking at explainable planning (requiring the agent to justify its steps) to build trust.

7.4. Cross-domain, benchmark diversity

Existing critical thinking benchmarks are narrow. We need more diverse test cases that are complex and reflective of many real-world environments, including multi-step tasks that are situated in social or physical environments. Future benchmarks should analyze not just the acquisition of the goal but also safety, robustness to changes, and ethical behavior. TheAgentCompany suggests the occupational domain as one place we could benchmark the agent’s behaviors; similarly, other contextual domains could be studied within healthcare, or smart cities, or even joint, creative endeavors [3].

7.5. Ethical AI frameworks

Lastly, it is urgent to connect the technical advances we have made with ethics. The only way to protect society moving forward will be for AI researchers, ethicists, and lawmakers to work together to create standards (privacy, liability, transparency, avoidance of bias) for agentic systems. We are just beginning the conversation on the legality around AI agency and what liability frameworks look like [6]. One way forward may also be to track the autonomy scales of agents to clarify their potential harms, akin to “certification” criteria we normally apply for testing in the production of pharmaceuticals. away.

8. Conclusion

Agentic AI is on the cutting edge of AI research, combining ideas about AI agents from the early 1990s with the power of large language models (LLMs) of today. The history of agentic AI suggests that we have moved from simple, rule-based agents to open-ended, self-directed agents as we have more ability to build LLMs as flexible planners and knowledge sources with memory and other external tools, which allow us to complete more complex tasks. While earlier applications of agentic AI for coding, science, robotics, and business are evidence of the possibility for transformational change, benchmarks indicate that agentic AI as we know it is still intermediary to fully autonomous agent behavior. As agentic AI continues developing, the new generation of AI will include three central technical and ethical concerns: 1) how can we ensure alignment with the user's intent? 2) how can we enable safe human-in-the-loop control? and 3) how can we understand the opaque and unforeseen decisions? These challenges will involve new modes of theorizing (agency and alignment), new standards for evaluation, and collaboration among diverse disciplines. In summary, agentic AI is a meaningful leap toward AI systems that do not just perform tasks but think and act more independently. Overall, the field is in a rapidly evolving stage - a lot has been made, but we are only just getting started toward AI that is fully autonomous and beneficial.

9. Conflict of Interest

The authors have no conflicts of interest to declare.

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